

## Unit-2

**TOPIC:-** Define Sensor, Basic components and challenges of a sensor node.

**OBJECTIVES:-** Sensors play an important role in creating solutions using IoT. Sensors are devices that detect external information, replacing it with a signal that humans and machines can distinguish.

**OUTCOME:-** To conclude, IoT has the capability to create public value and the intended public value dimensions have widened from the goal of improved efficiency.

### Sensors

Sensors science and engineering is relevant to virtually all aspects of life including safety, security, surveillance, monitoring, and awareness in general. Sensors are central to industrial applications being used for process control, monitoring, and safety. Sensors are also central to medicine being used for diagnostics, monitoring, critical care, and public health.

### Today's applications

What is exciting in sensor research and development today? That is a tough question. There are many significant innovations and inventions begin made daily. Micro- and nanotechnology, novel materials, and smaller, smarter, and more effective electronic systems will play an important role in the future of sensors.

To fulfill the promise of ubiquitous sensor systems providing situational awareness at low cost, there must be a demonstrated benefit that is only gained through further miniaturization. For example, new nanowire-based materials that have unique sensing properties can provide higher sensitivity, greater selectivity, and possibly improved stability at a lower cost. Such improvements are necessary to the sensor future.

### Future of sensor technology

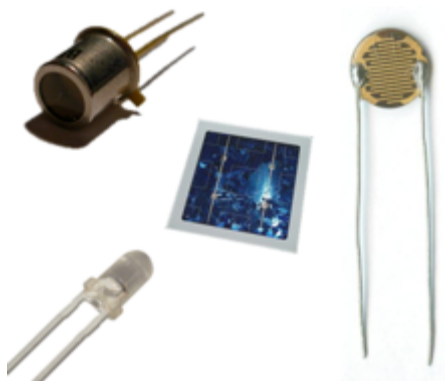
Sensors can improve the world through diagnostics in medical applications; improved performance of energy sources like fuel cells and batteries and solar power; improved health and safety and security for people; sensors for exploring space and the known universe; and improved environmental monitoring.

The seed technologies are now being developed for long-term vision that includes intelligent systems that are self-monitoring, self-correcting and repairing, and self-modifying or morphing not unlike sentient beings. The ability for a system to see (photonic technology), feel (physical measurements), smell (electronic noses), hear (ultrasonics), think/communicate (smart electronics and wireless), and move (sensors integrated with actuators), is progressing rapidly and suggests an exciting future for sensors.

In the broadest definition, a **sensor** is a device, module, machine, or subsystem whose purpose is to detect events or changes in its environment and send the information to other electronics, frequently a computer processor . A sensor is always used with other electronics.

Sensors are used in everyday objects such as touch-sensitive elevator buttons (tactile sensor ) and lamps which dim or brighten by touching the base, besides innumerable applications of which most people are never aware. With advances in micromachinery and easy-to-use microcontroller platforms, the uses of sensors have expanded beyond the traditional fields of temperature, pressure or flow measurement,

for example into MARG sensors . Moreover, analog sensors such as potentiometers and force-sensing resistors are still widely used. Applications include manufacturing and machinery, airplanes and aerospace, cars, medicine, robotics and many other aspects of our day-to-day life. There are a wide range of other sensors, measuring chemical & physical properties of materials. A few examples include optical sensors for Refractive index measurement, vibrational sensors for fluid viscosity measurement and electro-chemical sensor for monitoring pH of fluids.



Different types of light sensors

### **Classification of measurement errors**

A good sensor obeys the following rules:

- it is sensitive to the measured property

- it is insensitive to any other property likely to be encountered in its application, and
- it does not influence the measured property.

## Different Types of Sensors

We live in a World of Sensors. You can find different types of Sensors in our homes, offices, cars etc. working to make our lives easier by turning on the lights by detecting our presence, adjusting the room temperature, detect smoke or fire, make us delicious coffee, open garage doors as soon as our car is near the door and many other tasks.

All these and many other automation tasks are possible because of Sensors. Before going in to the details of What is a Sensor, What are the Different Types of Sensors and Applications of these different types of Sensors, we will first take a look at a simple example of an automated system, which is possible because of Sensors (and many other components as well).

## Real Time Application of Sensors

The example we are talking about here is the Autopilot System in aircrafts. Almost all civilian and military aircrafts have the feature of Automatic Flight Control system or sometimes called as Autopilot.

An Automatic Flight Control System consists of several sensors for various tasks like speed control, height, position, doors, obstacle, fuel, maneuvering and many more. A Computer takes data from all these sensors and processes them by comparing them with pre-designed values.

The computer then provides control signal to different parts like engines, flaps, rudders etc. that help in a smooth flight. The combination of Sensors, Computers and Mechanics makes it possible to run the plane in Autopilot Mode.

All the parameters i.e. the Sensors (which give inputs to the Computers), the Computers (the brains of the system) and the mechanics (the outputs of the system like engines and motors) are equally important in building a successful automated system.

But in this tutorial, we will be concentrating on the Sensors part of a system and look at different concepts associated with Sensors (like types, characteristics, classification etc.).

## Sensor

There are numerous definitions as to what a sensor is but I would like to define a Sensor as an input device which provides an output (signal) with respect to a specific physical quantity (input).

The term “input device” in the definition of a Sensor means that it is part of a bigger system which provides input to a main control system (like a Processor or a Microcontroller).

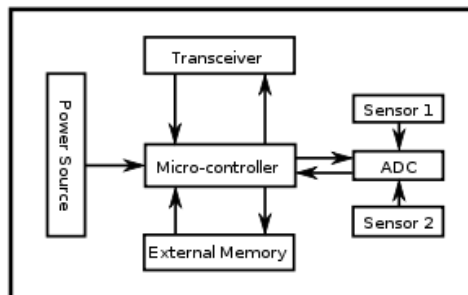
Another unique definition of a Sensor is as follows: It is a device that converts signals from one energy domain to electrical domain. The definition of the Sensor can be understood if we take an example in to consideration.

The simplest example of a sensor is an LDR or a Light Dependent Resistor. It is a device, whose resistance varies according to intensity of light it is subjected to. When the light falling on an LDR is more, its resistance becomes very less and when the light is less, well, the resistance of the LDR becomes very high.

We can connect this LDR in a voltage divider (along with other resistor) and check the voltage drop across the LDR. This voltage can be calibrated to the amount of light falling on the LDR. Hence, a Light Sensor.

Now that we have seen what a sensor is, we will proceed further with the classification of Sensors.

### Basic components of sensor



A **sensor node**, also known as a **mote** (chiefly in North America ), is a node in a sensor network that is capable of performing some processing, gathering sensory information and communicating with other connected nodes in the network. A mote is a node but a node is not always a mote.

### Controller

The controller performs tasks, processes data and controls the

functionality of other components in the sensor node. While the most common controller is a microcontroller , other alternatives that can be used as a controller are: a general purpose desktop microprocessor , digital signal processors , FPGAs and ASICs . A microcontroller is often used in many embedded systems such as sensor nodes because of its low cost, flexibility to connect to other devices, ease of programming, and low power consumption.

## **Transceiver**

Sensor nodes often make use of ISM band , which gives free radio , spectrum allocation and global availability. The possible choices of wireless transmission media are radio frequency (RF), optical communication (laser) and infrared . Lasers require less energy , but need line-of-sight for communication and are sensitive to atmospheric conditions. Infrared, like lasers, needs no antenna but it is limited in its broadcasting capacity.

## **External memory**

From an energy perspective, the most relevant kinds of memory are the on-chip memory of a microcontroller and Flash memory —off-chip RAM is rarely, if ever, used.

## **Power source**

A wireless sensor node is a popular solution when it is difficult or impossible to run a mains supply to the sensor node. However, since the wireless sensor node is often placed in a hard-to-reach location, changing the battery regularly can be costly and inconvenient.

## **Sensors**

Sensors are used by wireless sensor nodes to capture data from their environment. They are hardware devices that produce a measurable response to a change in a physical condition like temperature or pressure.

## **Classification of Sensors**

There are several classifications of sensors made by different authors and experts. Some are very simple and some are very complex. The following classification of sensors may already be used by an expert in the subject but this is a very simple classification of sensors.

In the first classification of the sensors, they are divided into Active and Passive. Active Sensors are those which require an external excitation signal or a power signal.

Passive Sensors, on the other hand, do not require any external power signal and directly generate output response.

The other type of classification is based on the means of detection used in the sensor. Some of the means of detection are Electric, Biological, Chemical, Radioactive etc.

The next classification is based on conversion phenomenon i.e. the input and the output. Some of the common conversion phenomena are Photoelectric, Thermoelectric, Electrochemical, Electromagnetic, Thermo-optic, etc.

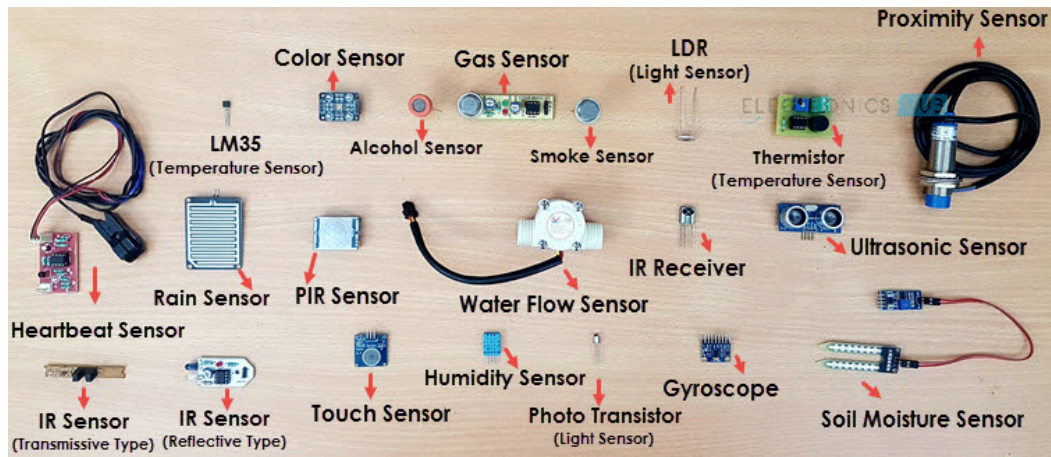
The final classification of the sensors are Analog and Digital Sensors. Analog Sensors produce an analog output i.e. a continuous output signal with respect to the quantity being measured.

Digital Sensors, in contrast to Analog Sensors, work with discrete or digital data. The data in digital sensors, which is used for conversion and transmission, is digital in nature.

### **Different Types of Sensors**

The following is a list of different types of sensors that are commonly used in various applications. All these sensors are used for measuring one of the physical properties like Temperature, Resistance, Capacitance, Conduction, Heat Transfer etc.

- Temperature Sensor
- Proximity Sensor
- Accelerometer
- IR Sensor (Infrared Sensor)
- Pressure Sensor
- Light Sensor
- Ultrasonic Sensor
- Smoke, Gas and Alcohol Sensor
- Touch Sensor
- Color Sensor
- Humidity Sensor
- Tilt Sensor
- Flow and Level Sensor



**Challenges** in such **WSN** include high bandwidth demand, high energy consumption, quality of service (QoS) provisioning, data processing and compressing techniques, and cross-layer design. physical environment. Presently, UWSNs face **issues** and **challenges** regarding limited bandwidth, high propagation delay, 3D topology, media access control, routing, resource utilization, and power **constraints**.

## REFERENCES:-

**SUMMARY:-** A sensor is a **device that detects and responds to some type of input from the physical environment**. The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.

**TOPIC:-** Sensor features, Sensor resolution.

**OBJECTIVES:-** Important static characteristics of sensors include **sensitivity, resolution, linearity, zero drift and full-scale drift, range, repeatability and reproducibility**. Sensitivity is a measure of the change in output of the sensor relative to a unit change in the input (the measured quantity.)

**OUTCOME:-** The sensor resolution or measurement resolution is **the smallest change that can be detected in the quantity that it is being measured**. The resolution of a sensor with a digital output is usually the numerical resolution of the digital output.

### **Sensor features, Sensor resolution**

Sensors are characterized depending on the value of some of the parameters. Important **characteristics of sensors** and transducers are listed below:

- Input characteristics
- Transfer characteristics
- Output characteristics

### **Input Characteristics of Sensors**

1. **Range:** It is the minimum and maximum value of physical variable that the sensor can sense or measure. For example, a Resistance Temperature Detector (RTD) for the measurement of temperature has a range of -200 to 800°C.
2. **Span:** It is the difference between the maximum and minimum values of input. In above example, the span of RTD is  $800 - (-200) = 1000^{\circ}\text{C}$ .
3. **Accuracy:** The error in measurement is specified in terms of accuracy. It is defined as the difference between measured value and true value. It is defined in terms of % of full scale or % of reading.

$$\text{Absolute Error} = |\text{Measured value} - \text{True value}|$$

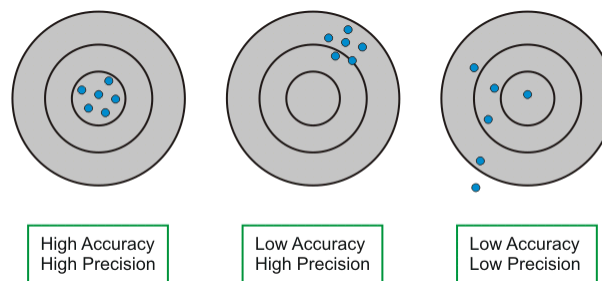
$$\Rightarrow E_a = |X_m - X_t|$$

$X_t$  is calculated by taking mean of infinite number of measurements.

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{True value}}$$

$$\Rightarrow E_r = \frac{|X_m - X_t|}{X_t}$$

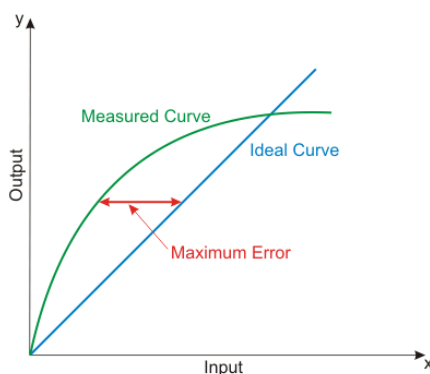
4. Precision: It is defined as the closeness among a set of values. It is different from accuracy. Let  $X_t$  be the true value of the variable  $X$  and a random experiment measures  $X_1, X_2, \dots, X_i$  as the value of  $X$ . We will say our measurements  $X_1, X_2, \dots, X_i$  are precise when they are very near to each other but not necessarily close to true value  $X_t$ . However, if we say  $X_1, X_2, \dots, X_i$  are accurate, it means that they are close to true value  $X_t$  and hence they are also close to each other. Hence accurate measurements are always precise.



5. Sensitivity: It is the ratio of change in output to change in input. If  $Y$  be the output quantity in response to input  $X$ , then sensitivity  $S$  can be expressed as

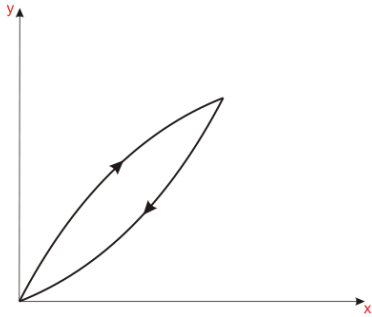
$$S = \frac{dY}{dX} = \frac{\Delta Y}{\Delta X}$$

6. Linearity: Linearity is the maximum deviation between the measured values of a sensor from ideal curve.



7. Hysteresis: It is the difference in output when input is varied in two

ways- increasing and decreasing.



8. Resolution: It is the minimum change in input that can be sensed by the sensor .
9. Reproducibility: It is defined as the ability of sensor to produce the same output when same input is applied.
10. Repeatability: It is defined as the ability of sensor to produce the same output every time when the same input is applied and all the physical and measurement conditions kept the same including the operator, instrument, ambient conditions etc.
11. Response Time: It is generally expressed as the time at which the output reaches a certain percentage (for instance, 95%) of its final value, in response to a step change of the input.

## Sensor Resolution

The resolution of remote sensed raster data can be characterized in several different ways. There are four primary types of "resolution" for rasters:

- Spatial
- Spectral
- Radiometric
- Temporal

It is nearly impossible to acquire imagery that has high spatial, spectral, radiometric and temporal resolution. This is known as **Resolution Trade-off**, as it is difficult and expensive to obtain imagery with extremely high resolution. Therefore it is necessary to identify which types of resolution are most important for a project.

### Spatial

Spatial resolution is the type of resolution most people are familiar with. Spatial resolution is usually reported as the length of one side of a single pixel. For example, Landsat 8 has 30 meter spatial resolution. In other

words, an image with 30 meter spatial resolution means that a single pixel represents an area on the ground that is 30 meters across.

In analog imagery (film), the dimension (or width) of the smallest object on the ground that can be distinguished in the imagery defines the spatial resolution. The spatial resolution of a raster is determined by sensor characteristics for digital imagery and film characteristics including field of view, altitude for film photography.

#### Satellite Data Spatial Resolution

##### **30m Resolution 15m Resolution 1m Resolution**

| <b>Sensor</b>         | <b>Spatial Resolution</b> |
|-----------------------|---------------------------|
| MODIS                 | 250-1000 m                |
| Landsat MSS           | 60 m                      |
| Landsat TM, ETM+, OLI | 30 m                      |
| Sentinel-2            | 10-60 m                   |
| ASTER                 | 15 m                      |
| Digital Globe         | 0.3 m (30 cm)             |

The higher the resolution of the image, the more expensive it is to capture, process, and distribute. Part of this is just the size of the image. As the spatial resolution increases, the size of the raster goes up by the increase in resolution squared! This makes spatial resolution critical to control as we need enough resolution to see what we are interested in on the ground but not so much that we cannot process the data effectively. The spatial resolution of sensors varies greatly and is usually inversely related to return time and coverage area. This means that a satellite that acquires daily images generally has a more coarse (larger) spatial resolution.

## Spectral

Spectral resolution refers to how many spectral “bands” an instrument records. Spectral resolution is also defined by how “wide” each band is or the range of wavelengths covered by a single band. Black and white photos contain only 1 band that covers the visible wavelengths, color (RGB ) images contain 3 bands and Landsat 8 has a total of 11 bands. For example, MODIS (Moderate Resolution Imaging Spectroradiometer) has greater spectral resolution than Landsat 8 because it has 36 relatively narrow bands that cover wavelengths from 0.4 to 14 micrometers. Landsat 8, on the other hand, has a total of 11 bands that cover less wavelengths and each band is wider in terms of wavelength.

## Radiometric

Radiometric resolution is how finely a satellite or sensor divides up the radiance it receives in each band. The greater the radiometric resolution the greater the range of intensities of radiation the sensor is able to distinguish and record. Radiometric resolution is typically expressed as the number of bits for each band. Traditionally 8-bit data was common in remote sensed data, newer sensors (like Landsat 8) have 16-bit data products.  $8 \text{ bits} = 2^8 = 256 \text{ levels}$  (usually 0 to 255)  $16 \text{ bits} = 2^{16} = 65,536 \text{ levels}$  (0 to 65,535).

| 8 bit Resolution           | 2 bit Resolution         | 1 bit Resolution         |
|----------------------------|--------------------------|--------------------------|
| $2^8 = 256 \text{ levels}$ | $2^2 = 4 \text{ levels}$ | $2^1 = 2 \text{ levels}$ |

## Temporal

Remote sensed data represents a snap shot in time. Temporal resolution is the time between two subsequent data acquisitions for an area. This is also known as the “return time” or “revisit time”. The temporal resolution depends primarily on the platform, for example, satellites usually have set return times and while sensors mounted on aircraft or unmanned aircraft systems ( UAS ), have variable return times. For satellites, the return time depends on the orbital characteristics (low vs high orbit), the swath width and whether or not there is an ability to point the sensor. Landsat has a return time of approximately 16 days, while other sensors like MODIS have nearly daily return times.

## **REFERENCES:-**

## **SUMMARY:-**

The sensor resolution or measurement resolution is the smallest change that can be detected in the quantity that it is being measured.

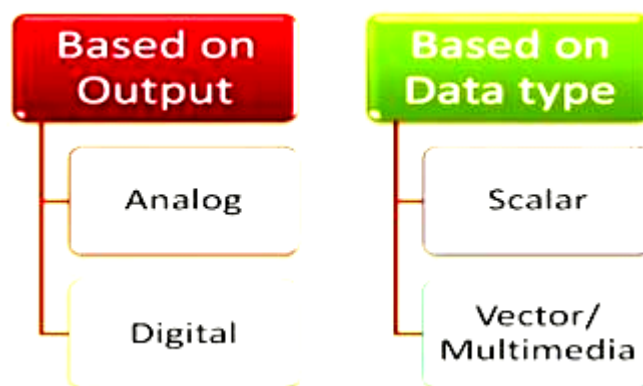
**TOPIC:-** Sensor classes: Analog, Digital, Scalar, Vector Sensors

**OBJECTIVES:-** Analog sensors create analog signals from the quantity that is sensed. In comparison, digital sensors produce digital signals when performing a measurement.

**OUTCOME:-** To understand about most common types of analog sensors include sound sensors, light sensors, temperature sensors, and pressure sensors.

Sensor classes:

As we all are living in digital technology, the one device that became an integral part of our daily activity is Sensor. Every device which we come across in our home from mobile phone to security systems at home, all are embedded with various kind of sensors. So, a sensor is a device that observes changes occurring in the parameters such as humidity, pressure, temperature, and others and sends the output to either microcontroller or microprocessor.



### Analog Sensors

Analog sensors are the devices that produce analog output in correspondence to the quantity being calculated. These sensors also observe the change in external factors such as light intensity, speed of the wind, and solar radiation, and others. And the output ranges between 0V to 5V. Here, Logic 1 corresponds to the voltage level of 3.5V to 5V and Logic 0 corresponds to the voltage level of 0V to 3.5V.

For instance, the temperature range of a liquid can be known either by thermocouple or thermometer that constantly responds to temperature change either the liquid is cooled or heated.

## Types of Analog Sensors

Analog sensors are classified as many types and they are of:

- Accelerometers
- Light sensors
- Sound sensors
- Pressure sensors
- Analog temperature sensors

Let's have a brief discussion on all these types of analog sensors

### *Accelerometers*

The name itself indicates that these are the type of analog sensors where those can be utilized to detect changes in acceleration applied on the sensor. Acceleration also corresponds to the change in the values of velocity, vibration, position, and tilt by detecting motion. These accelerometers deliver two kinds of data. The static force exerted on the sensor because of gravity that results to know the change in orientation.

- Force exerted on the sensor results in a change in the movement.

The foremost benefit of implementing the accelerometer sensor is biometrics. The general usage of accelerometer sensors is in vibration motion and in screen orientational changes. But the drawback in these devices is inter-device variations.

### *Light Sensors*

These are the kind of photoelectric devices where the detected light energy is converted to that of electrical energy which means that photons are converted as electrons. Light sensors are mostly helpful to find out and react for various light levels such as in machines, switches, and other appliances. These are further classified depending on the collected current or hold voltage based on the levels of light. Mostly, light sensors are implemented for robotic intelligence and motion lights. Even these devices hold the ability to detect light rays not visible for humans such as UV rays, infrared, and X-rays.



light-analog-sensor

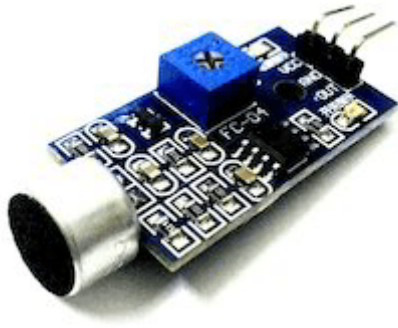
The types of light sensors are:

- **Photovoltaic** – These are also termed as solar cells. These sensors act in response to light levels either by generating voltage or current and stored in silicon cells for further usage. These are minimal and produce only minimal wattages and for high currents, these will be available as panels.
- **Light-dependent** – This type of light sensor is mainly utilized for gauging purposes and reacting for light levels. They operate as automatic switches for various devices. These devices come under the classification of photoresistors as the reason their resistance value enhances as the light levels go up.
- **Photo Diode** – Devices such as remote controls, cameras, and video recorders utilize photodiodes to identify light levels. This device acts in response to various infrared levels and performs as switches. In general, even for minimal light changes, these devices deliver minimal current.
- **Proximity** – In correspondence to the infrared light changes, proximity sensors detect either the closeness of another item or detects motion. These are helpful in robotic machines to pass through obstacles and keep away from object bumpings.

### *Sound Sensors*

Sound sensors are considered as the modules to find out sound waves by the intensity and converting those into electrical signals. It means that the amplitude levels of the sound wave are converted to electrical voltage. Sound sensors are used in the applications of door alarms, music systems,

burglar alarms, computers, and many others.



sound-sensor

### *Pressure Sensor*

Pressure sensors are the devices used for the measurement of pressure levels of either liquids or gases. This device is even utilized to measure a few of the other parameters like the altitude, flow of either gas or liquid, and water levels. Pressure sensors are also termed as piezometers, pressure transmitters, pressure indicators, and manometers.

Few of the pressure sensors also operated as pressure switches, which gets ON or OFF for corresponding pressure levels. For instance, the water pump can be handled through pressure switch in the way when the switch is in 'ON' condition, water will be released thus decreasing the pressure levels in the reservoir. These are classified as

- Absolute pressure sensors
- Differential pressure sensors
- Gauge sensors
- Sealed gauge sensors

### *Analog Temperature Sensor*

These sensors deliver analog output voltage or current signals in linear to the temperature levels. The usage of these devices is so simple, and no need for complicated circuits to construct. The accuracy of output ranges from  $+1^{\circ}\text{C}$  to  $-1^{\circ}\text{C}$ . These are also called as thermistors. when the temperature rate enhances, then the value of electrical resistance also gets increased. In the same way, if the temperature rate is reduced, then the resistance value declines. Also, low power usage of these analog temperature sensors reduces the power utilization of a system, self-heating, and even the construction is also so simple.

## Applications of Analog Sensors

As there are various classifications in analog sensors, there is an extensive range of applications in many industries of these devices. Few of the applications are:

- Parking sensors
- Object alignments
- Mobile phones
- Industrial fabrications
- Automobiles
- Robotics

## Analog signal

An analog signal generally corresponds to linearity. Analog systems are utilized when the output levels of either current or voltage have to be proportional as per input values. These systems have unlimited variability in nature.

## Benefits of analog signals

Analog signals have less distortion and less quantization noise. And also, extensive performance even in high load conditions.

## Advantages of analog sensors

Analog sensors are the devices that produce analog output in correspondence to the quantity being calculated. The quantity which is measured can be temperature, pressure, force, light, and others.

## Sensors that produce analog output

Those are flex sensors, temperature sensors, UV light sensors, microphones, force-sensitive sensors, and photocells.

## Phone digital or analog

A phone operates on both analog and digital signals. Analog signals are used for transmission and reception of communication signals whereas the data inside the analog is digital in nature.

## Digital Sensors

Electrochemical and electrical sensors are usually one part of the measuring sequence. And this sequence generally comprises of a sensor, transmitter, and a cable. And the foremost classification of sensors is

analog and digital. The digital sensor technology is more substantial in regard to its extensive reliability and exactness.

Digital sensors are the kind of electrochemical or electrical sensors where the information is converted to digital form and then transmitted. The output of a digital sensor is the distinct digital signal of the quantity which is being measured. And the measured quantities might be of conductivity, pH value, redox potentials, and many others. The output is in the form of 1's and 0's where '1' represents ON condition and '0' represents OFF condition. This corresponds that a digital signal generates distinct (non-continuous) values and the output is considered either as a single "bit", (serial transmission) or the combination of multiple bits called "byte" and is called (parallel transmission).

### Types of Digital Sensors

Digital sensors are also classified as many types and few of those to be described as follows:

- Digital accelerometers
- Digital temperature sensor

#### *Digital Accelerometer*

Digital accelerometers generally make use of PWM (pulse width modulation) to generate output pulses. It corresponds that the output is a square wave of corresponding frequency and the time for which the voltage is high is linear to the amount of time taken for acceleration.

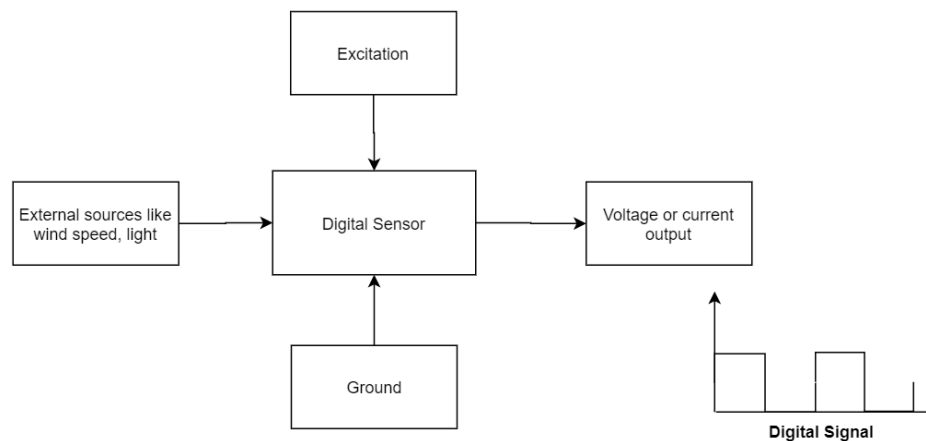
When a BASIC stamp or any kind of microcontroller is used as digital inputs, then the digital accelerometer is the preferred one to generate output..

Electrochemical and electrical sensors are usually one part of the measuring sequence. And this sequence generally comprises of a sensor, transmitter, and a cable. And the foremost classification of sensors is analog and digital. The digital sensor technology is more substantial in regard to its extensive reliability and exactness.

### Digital Sensors

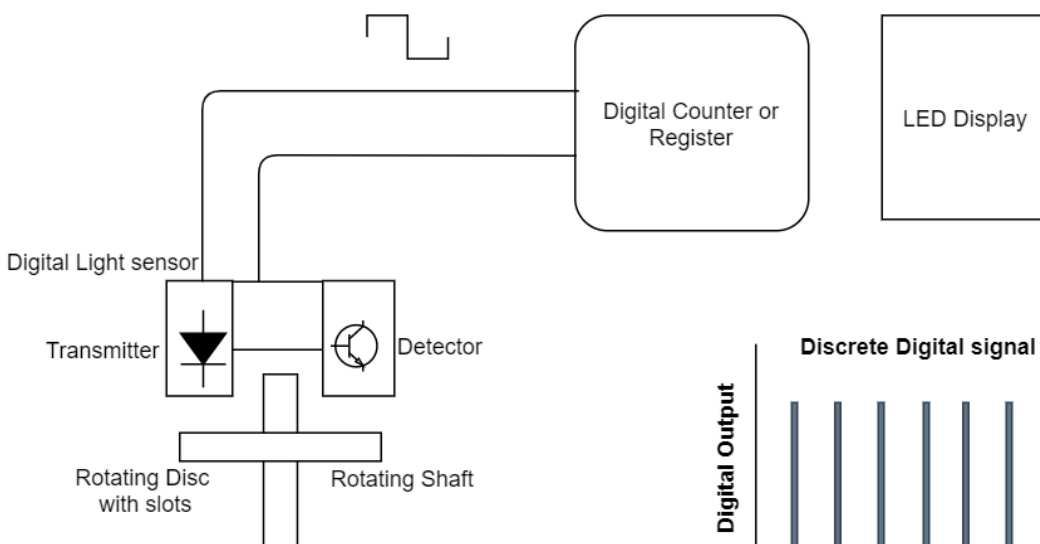
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## Basic Digital Sensor

In the diagram, the rotating shaft speed is known with the help of a digital LED light sensor. Here, the place of the disk is positioned to the revolving shaft which has transparent slots internal to the disk design. As both the shaft and disc rotational speed are the same, every slot in the disk passes through the sensor thus generating output pulses in the form of logic '1' and logic '0' (the digital signal).



## Digital Sensor Output

The generated pulses are then transmitted to the register in the counter and then to the output which displays the number of shaft revolutions. The number of output pulses for every rotation can be increased by augmenting the number of disk slots. The benefit of this increment is that higher accurateness and resolution can be obtained and minimal revolutions can also be observed easily. So, this kind of sensor construction is mainly utilized for controlling positions of the slots in the disk where this acts as a reference position.

So, in comparison to analog signals, discrete signals are with extensive accurateness and they can be calculated even at increased clock speeds. The exactness of the digital pulse is linear to the bits that are utilized for quantity measurement.

For instance, a processor of 8 bits, will generate an accuracy of nearly 0.390 percent, whereas with a 16-bit processor and accuracy of almost 0.0015 percent can be achieved. So, this accurateness can be asserted as digital quantities and these are many times quicker than that of analog signals.

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When a BASIC stamp or any kind of microcontroller is used as digital inputs, then the digital accelerometer is the preferred one to generate output.

### Features of Digital Accelerometers

- The data rates can be selected by users
- Operates using FIFO/FILO memory buffers
- One can achieve digital high-pass filter outputs
- Minimal power consumption
- Solderability can be done without using lead

- Exceptional temperature performance
- Extensive shock survivability
- Factory programmable offset and compassion

## **Applications**

- Implemented in mobile phones and other internet equipment's
- Used in-game controllers and computer components
- Used in the health care industry
- Also implemented in personal navigation equipment

## *Digital Temperature Sensors*

These are silicon dependent temperature sensors where the output is the accurate digital representation of the measured temperatures. These devices are designed to read the temperature ranges from  $0^{\circ}\text{C}$  to that of  $70^{\circ}\text{C}$  and with this, an output of nearly  $\pm 0.5^{\circ}\text{C}$  accurateness can be achieved. Whereas, packaged components are designed for extensive ranges which means for  $-55^{\circ}\text{C}$  to  $175^{\circ}\text{C}$ , the accuracy is  $\pm 1^{\circ}\text{C}$  and for  $-130^{\circ}\text{C}$  to  $150^{\circ}\text{C}$ , the accuracy is  $\pm 1.5^{\circ}\text{C}$ .

## **The flexibility of Digital Temperature Sensors**

In general, registers are programmed with the temperature levels of high, low, and medium. The calculated value is related to that of these temperature limits and either increased or decreased values apart from these limits prompts corresponding pins on the package. So, through the serial bus, the calculated temperature value is transmitted to the system controller which is even utilized for configuration.

On the other hand, these digital temperature sensors operate basically as thermostats.

## **Example:**

DS18B20 is a kind of single wired digital temperature sensor. Each of these sensors is wit a 64-bit serial number engraved into it which permits for an extensive number of sensors to be utilized on a single data bus. The features of this sensor are as follows:

- This singled wired sensor needs only one port for communication purposes.
- The feature called multidrop ability streamlines distributed

temperature sensing functionalities.

- No additional components are required
- It can be powered up through data lines
- The operating power supply ranges from 3.0V to 5.5V
- It can measure temperatures ranging between  $-55^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$  providing an accuracy of  $-10^{\circ}\text{C}$  to  $+85^{\circ}$
- The resolution of a thermometer is customizable selecting from 9 -12 bits.
- Through alarm search command detects and addresses the devices where the temperature range is external to that of programmed limits.

With the one can measure the temperature ranges of air, water, and even the ground temperature.



Digital sensors

### Use of digital sensors

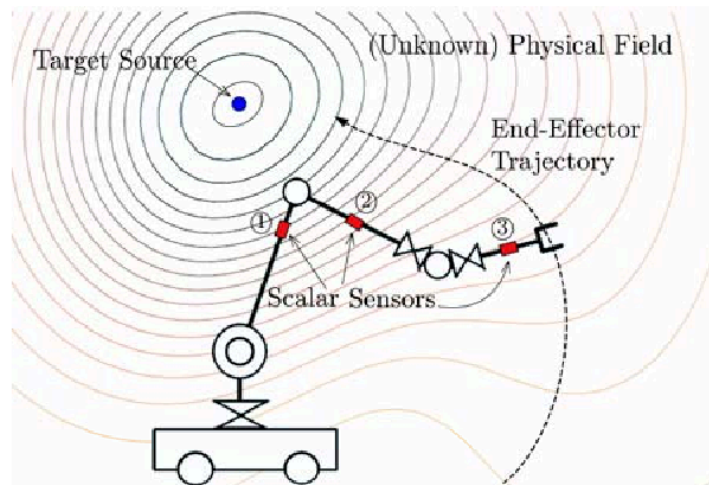
- Measuring signals are simply converted to digital signal internal to the sensor
- Digital sensors have no complication for humidity and corrosion
- These can be calibrated separately from the system

### Scalar sensor

- The sensor which produce output signal/voltage which is proportional to magnitude of quantity to be measured is known as scalar sensor.
- Temperature of a room is measured using scalar sensor (i.e. thermometer or thermocouple) which measures temperature changes irrespective of sensor orientation.

### • Examples:

Temperature sensor, Color sensor, pressure sensor, strain sensor etc.



## Vector sensor

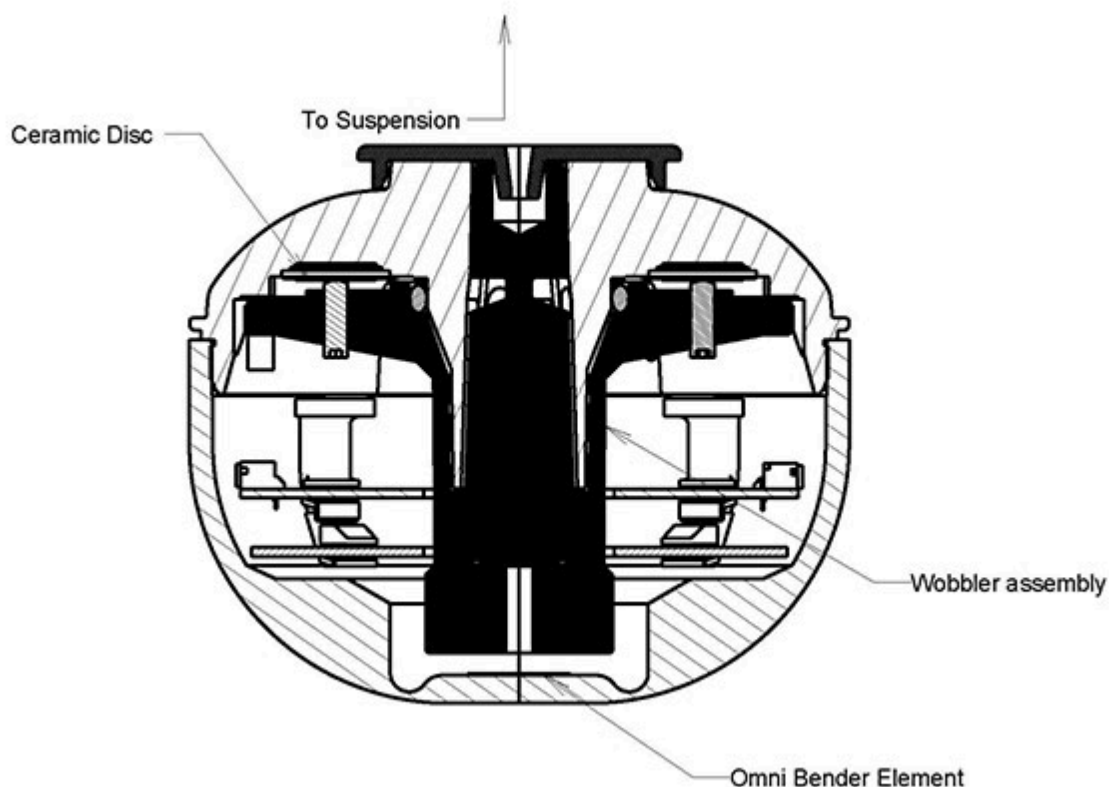
- The sensor which produce output signal/voltage which is proportional to magnitude, direction as well as orientation of quantity being measured is known as Vector sensor.
- Acceleration of body is measured using accelerometer which measures component of acceleration of body with respect to x, y, z coordinate axes.
- Examples:  
Sound sensor, image sensor, velocity sensor, acceleration sensor etc.

➤ The major difference between scalar sensor and vector sensor is based on type of data to be measured. As mentioned above, scalar sensor measures scalar data where as vector sensor measures vector data.

## Vector Sensors

A vector sensor is an underwater listening device used to detect sounds in water and convert acoustic energy into electrical energy. Vector sensors differ from hydrophones, which are also underwater listening devices, in that vector sensors measure both the particle motion and pressure changes associated with a sound wave, while hydrophones measure only the pressure changes. Vector sensors can determine the direction in which a sound wave is traveling using measurements made at a single point in space. Individual hydrophones are not sensitive to direction, making it necessary to construct hydrophone arrays made up of a number of hydrophones to determine the direction in which a sound wave is traveling. The particle motion in a sound wave is described by displacement, velocity, and acceleration. These three quantities are vectors, because each has a direction as well as a magnitude. Vector sensors measure one of these vector quantities together with pressure. (Some vector sensors actually measure pressure gradient, which is proportional to acceleration, rather than particle motion itself.) For example,

a vector sensor can be created by appropriately combining two geophones, which generate electric signals proportional to the particle velocity in a sound wave, and a standard hydrophone, which measures pressure. While hydrophones are the most common underwater acoustic sensors, vector sensors have been used in U.S. Navy DIFAR (Directional Frequency Analysis and Recording) AN/SSQ-53 sonobuoys for many years. The DIFAR sensor consists of an omnidirectional hydrophone that measures pressure and two particle motion sensors mounted at right angles to each other that measure the horizontal components of particle motion associated with the sound wave. All three sensors are co-located in a single package that has nearly the same density as seawater so that the package moves with the sound wave in the same way as the surrounding water particles.



Schematic of a DIFAR sensor assembly showing the omnidirectional hydrophone ("Omni Bender Element") and two of the four arms of the "Wobbler Assembly" that forms the two horizontal accelerometers. Image credit: Ultra Electronics Maritime Systems.

## REFERENCES:-

**SUMMARY:-** All types of sensors are further divided into **analog and digital sensors**. However, there are few types of sensors that are used in electronic applications such as pressure sensors, touch sensors, IR sensors, ultrasonic sensors, temperature sensors, proximity sensors etc.

**TOPIC:-** Sensor Types, bias, drift, Hysteresis error, quantization error

**OBJECTIVES:-** The hysteresis error of a pressure sensor is the maximum difference in output at any measurement value within the sensor's specified range when approaching the point first with increasing and then with decreasing pressure.

**OUTCOME:-** By using the maximum working pressure, the accuracy is, of course, better than specified by the manufacturer (for example the percentage of working pressure). Also, the hysteresis error is usually expressed as a combination of mechanical and temperature hysteresis.

## **Sensor Types**

### **Temperature Sensor**

One of the most common and most popular sensor is the Temperature Sensor. A Temperature Sensor, as the name suggests, senses the temperature i.e. it measures the changes in the temperature.

In a Temperature Sensor, the changes in the Temperature correspond to change in its physical property like resistance or voltage.

There are different types of Temperature Sensors like Temperature Sensor ICs (like LM35), Thermistors, Thermocouples, RTD (Resistive Temperature Devices), etc.

Temperature Sensors are used everywhere like computers, mobile phones, automobiles, air conditioning systems, industries etc.

### **Proximity Sensors**

A Proximity Sensor is a non-contact type sensor that detects the presence of an object. Proximity Sensors can be implemented using different techniques like Optical (like Infrared or Laser), Ultrasonic, Hall Effect, Capacitive, etc.

Some of the applications of Proximity Sensors are Mobile Phones, Cars (Parking Sensors), industries (object alignment), Ground Proximity in Aircrafts, etc.

## **Infrared Sensor (IR Sensor)**

IR Sensors or Infrared Sensor are light based sensor that are used in various applications like Proximity and Object Detection. IR Sensors are used as proximity sensors in almost all mobile phones.

There are two types of Infrared or IR Sensors: Transmissive Type and Reflective Type. In Transmissive Type IR Sensor, the IR Transmitter (usually an IR LED) and the IR Detector (usually a Photo Diode) are positioned facing each other so that when an object passes between them, the sensor detects the object.

The other type of IR Sensor is a Reflective Type IR Sensor. In this, the transmitter and the detector are positioned adjacent to each other facing the object. When an object comes in front of the sensor, the sensor detects the object.

Different applications where IR Sensor is implemented are Mobile Phones, Robots, Industrial assembly, automobiles etc.

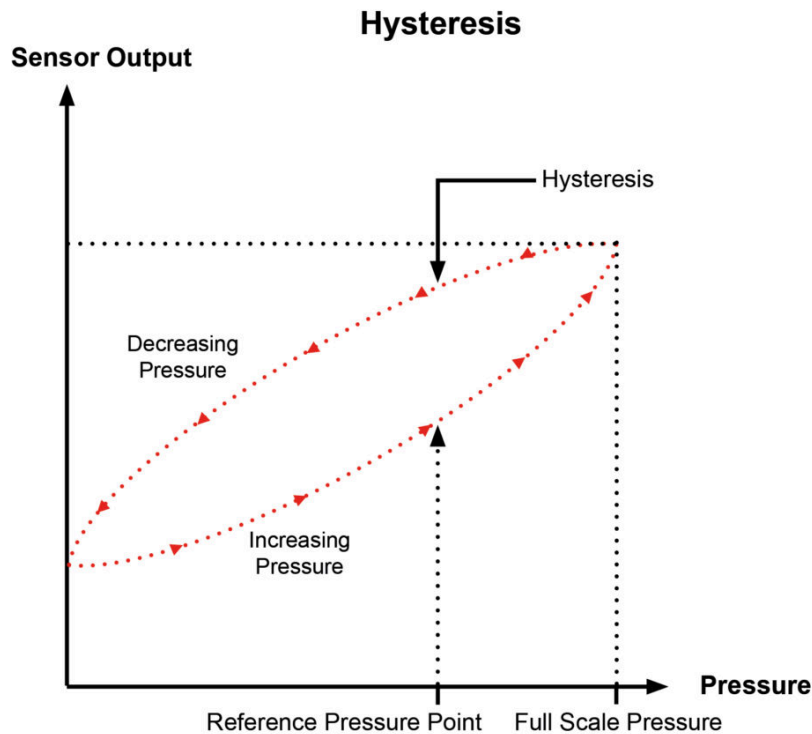
## **Ultrasonic Sensor**

An Ultrasonic Sensor is a non-contact type device that can be used to measure distance as well as velocity of an object. An Ultrasonic Sensor works based on the properties of the sound waves with frequency greater than that of the human audible range.

Using the time of flight of the sound wave, an Ultrasonic Sensor can measure the distance of the object (similar to SONAR). The Doppler Shift property of the sound wave is used to measure the velocity of an object.

## **Hysteresis Errors**

The hysteresis error of a pressure sensor is the maximum difference in output at any measurement value within the sensor's specified range when approaching the point first with increasing and then with decreasing pressure.



The hysteresis error value is normally specified as a positive or negative percentage of the specified pressure range. If a sensor is only used over half of the specified range the hysteresis error is calculated from this value.

By using the maximum working pressure, the accuracy is, of course, better than specified by the manufacturer (for example the percentage of working pressure). Also, the hysteresis error is usually expressed as a combination of mechanical and temperature hysteresis.

### Mechanical Hysteresis

Mechanical hysteresis is the output deviation at a certain input pressure when that input is approached first by increasing and then by decreasing pressure.

### Temperature Hysteresis

Temperature hysteresis is the output deviation at a certain input pressure, before and after a temperature cycle.

Hysteresis errors are not always specified separately but combined in a total figure for linearity, hysteresis and repeatability.

**Bias drift** is probably the most crucial of the errors. In inertial sensors even in the absence of any input (acceleration or angular velocity) the output is non-zero. This offset, which is usually referred to as **bias**, is added to the actual measured signal. During integration the errors of the measurements accumulate.

### Quantization in Digital Filters

Quantization errors in digital filters can be classified as:

- Round-off errors derived from internal signals that are quantized before or after more down additions;
- Deviations in the filter response due to finite word length representation of multiplier coefficients; and
- Errors due to representation of the input signal with a set of discrete levels.

### REFERENCES:-

### SUMMARY:-

Temperature hysteresis is the output deviation at a certain input pressure, before and after a temperature cycle. Hysteresis errors are not always specified separately but combined in a total figure for linearity, hysteresis and repeatability.

TOPIC:- **Actuator; Actuator types: Hydraulic, Pneumatic, electrical**

**OBJECTIVES:-** An actuator is a device that **produces a motion by converting energy and signals going into the system**. The motion it produces can be either rotary or linear. An actuator is a device that produces a motion by converting energy and signals going into the system.

**OUTCOME:-** An actuator is an output device for the **conversion of supply energy into useful work**. The output signal is controlled by the control system, and the actuator responds to the control signals via the control element.

### **Actuator; Actuator types: Hydraulic, Pneumatic, electrical**

An actuator is a device that uses a form of power to convert a control signal into mechanical motion. From electric door locks in automobiles, to ailerons on aircraft, actuators are all around us. Industrial plants use actuators to operate valves, dampers, fluid couplings, and other devices used in industrial process control . The industrial actuator can use air, hydraulic fluid, or electricity for motive power. These are referred to as pneumatic, electro-hydraulic, or electric actuators.

Actuators are mechanical or electro-mechanical devices that provide controlled and sometimes limited movements or positioning which are operated electrically, manually, or by various fluids such as air, hydraulic, etc. Two basic motions are linear and rotary. Linear actuators convert energy into straight line motions, typically for positioning applications, and usually have a push and pull function. Some linear actuators are unpowered and manually operated by use of a rotating knob or handwheel. Rotary actuators convert energy to provide rotary motion. A typical use is the control of various valves such as a ball valves or butterfly valves . Each actuator type has versions for various power configurations and come in many styles and sizes depending on the application. Linear chain actuators provide push and pull motions with rigid chains.



## Electro-mechanical actuators

### Types of Actuators

The various types of actuators.

- Electric Linear
- Electric Rotary
- Fluid Power Linear
- Fluid Power Rotary
- Linear Chain Actuators
- Manual Linear
- Manual Rotary

#### Electric Linear

Electric Linear Actuators are electrically powered, mechanical linear actuators consisting of motors, linear guides, and drive mechanisms, which are used to convert electrical energy into linear displacement through mechanical transmission, electro-magnetism, or thermal expansion to provide straight line push/pull motion. Key specifications include the intended application, drive type, motor type, the mounting configuration, as well as other physical dimensions and electrical characteristics. Electric linear actuators are used primarily in automation applications when a machine component, tool, etc. requires a controlled movement to a particular position. They are used in a wide range of industries where linear positioning is needed. The actuators are driven by several means including ball or lead screws, belts, or voice coils, among others. Typical applications include opening and closing dampers, locking doors, braking machine motions, etc.

#### Electric Rotary

Electric Rotary Actuators are electrically powered, mechanical devices consisting of motors and output shaft mechanisms with limited rotary travel which are used to convert electrical energy into rotational motion. Key specifications for these Rotary Actuators include the intended application, drive method, number of positions, output configuration, mounting configuration, as well as physical dimensions and electrical characteristics as required. Electric rotary actuators are used primarily in automation applications when a gate, valve, etc. requires controlled movement to particular rotational positions. They are used in a wide range of industries where positioning is needed. The actuators are driven by various motor types, voice coils, etc. Typical applications include quarter-

turn valves, windows, robotics, etc.

### **Fluid Power Linear Actuators**

Fluid Power Linear Actuators are mechanical devices consisting of cylinder and piston mechanisms that produce linear displacement by means of hydraulic fluid, gas, or differential air pressure. Key specifications include the intended application, fluid power type, mounting configuration, travel length, and force capacities, as well as physical dimensions as required. Fluid power linear actuators are used primarily in automation applications when an item requires a controlled movement to a particular position. They are used in a wide range of industries where linear positioning is needed. Typical applications include the opening and closing of damper doors, clamping, welding, etc.

### **Fluid Power Rotary Actuators**

Fluid Power Rotary Actuators are fluid powered, mechanical devices consisting of cylinder and piston mechanisms, gearing, and output shafts giving limited rotational travel, which are used to convert hydraulic fluid, gas, or differential air pressure into rotational motion. Key specifications include the intended application, fluid power type, drive mechanism, mounting configuration, output configuration, rotation limits, and force capacities, as well as physical dimensions as required. Fluid power rotary actuators are used primarily in automation applications when an item requires a controlled rotary movement to a particular position. They are used in a wide range of industries where rotational positioning is needed. The actuators are powered by various media including air or other gases and hydraulic fluid. Typical applications include opening and closing dampers and doors, clamping, etc.

### **Linear Chain Actuators**

Linear Chain Actuators are mechanical devices consisting of sprockets and sections of chain which are used for providing linear motion via the free ends of the specially designed chains. Key specifications include the intended application, drive method and mechanism, actuation length, chain size, and the mounting configuration. Linear chain actuators are used primarily in motion control applications for providing a straight line push or pull motion. The chain, when straight locks into position with adjacent links and forms a rigid member. They are available in many sizes and chain styles, as well as chain storage options depending on available space in a particular application. The actuators typically have single driving gears or sprockets that produce the forces required for the push and pull motions.

## **Manual Linear Actuators**

Manual Linear Actuators are mechanical devices providing linear displacement through the translation of manually rotated screws or gears and consist of hand-operated knobs or wheels, gearboxes, and guided linear motion mechanisms. Key specifications include the intended application, actuator type, drive mechanism, travel length, and other physical dimensions as required. Manual linear actuators are used primarily in commercial applications for precise positioning such as manipulating tools or workpieces. Manual actuators are unpowered and use a rotating knob or handwheel to operate. There are several types including those with lead screws, racks and pinions, belt drive, etc. each having its specific load and drive force capability. Thermo-mechanical actuators are another type and operate by means of thermal expansion, usually on a micro scale.

## **Manual Rotary Actuators**

Manual Rotary Actuators are mechanical devices providing rotary output through the translation of manually rotated screws, levers, or gears, and are usually composed of hand-operated knobs, levers, or handwheels, gearboxes or threaded nut mechanisms, and output shafts. Key specifications include the intended application, drive method and mechanism, mounting configuration, valve type, if applicable, as well as physical dimensions as required. Manual rotary actuators are used primarily for the operation of valves. They are also known as manual valve actuators or valve operators. Valve types can include ball, butterfly, check, and globe valves. Other applications may include any that require limited and controlled rotational movement.

## **Actuator Applications and Industries**

Powered actuators generally incorporate motors, cylinders, or other motion devices to produce rotation or linear translation. Rack and pinion arrangements, ball screws, and other kinds of power transmission components connect the motors, etc. to the loads.

Actuators are segregated by motion and power source. Linear actuators produce push/pull action. Rotary actuators produce rotational motion. In many cases, linear actuators begin with a rotary prime mover—a motor, typically—whose rotation is converted to linear motion through a power screw or similar device.

## **Everyday Examples**

In very simple terms, an actuator is a device that makes something move

or operate. Every one of us takes advantage of at least one actuator every day. Let's look at some examples of actuators.

### *1. Grocery Store Door*

When we go to the grocery store, the door opens automatically for us. An actuator makes the door open.



### *2. Car Seat*

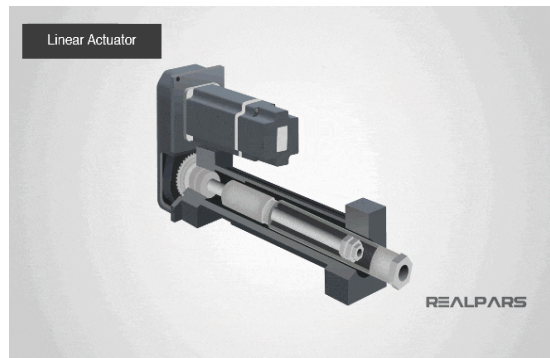
We can move the car seat forward or backward before we drive away in our car. An actuator makes the seat move.



## Types of Actuators

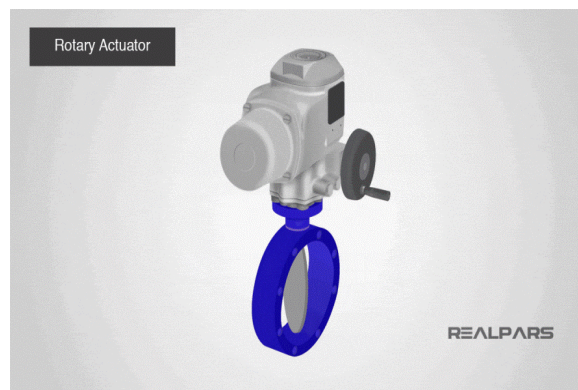
### *1. Linear Actuator*

An actuator can move something in a straight line, also referred to as **linear**.



## 2. Rotary Actuator

Also, An actuator can make something move in a circular motion, also referred to as **rotary**.



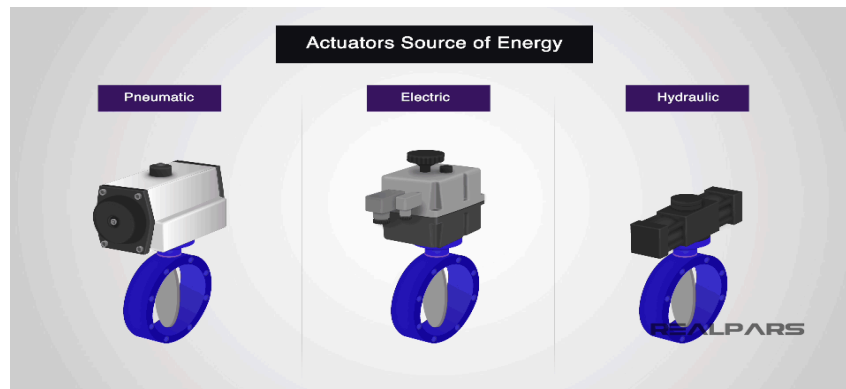
### What Makes an Actuator Move?

We've talked about the movement of an actuator, but we haven't discussed how, or what makes it move.

Actuators receive a source of energy and use it to move something. To put it another way, the actuator converts a source of energy into a physical-mechanical motion.

A butterfly valve is operated by turning the **Handwheel** connected to the rotary actuator. In industrial applications, the **Handwheel** could be replaced by one of three typical sources of energy, which can operate the rotary actuator:

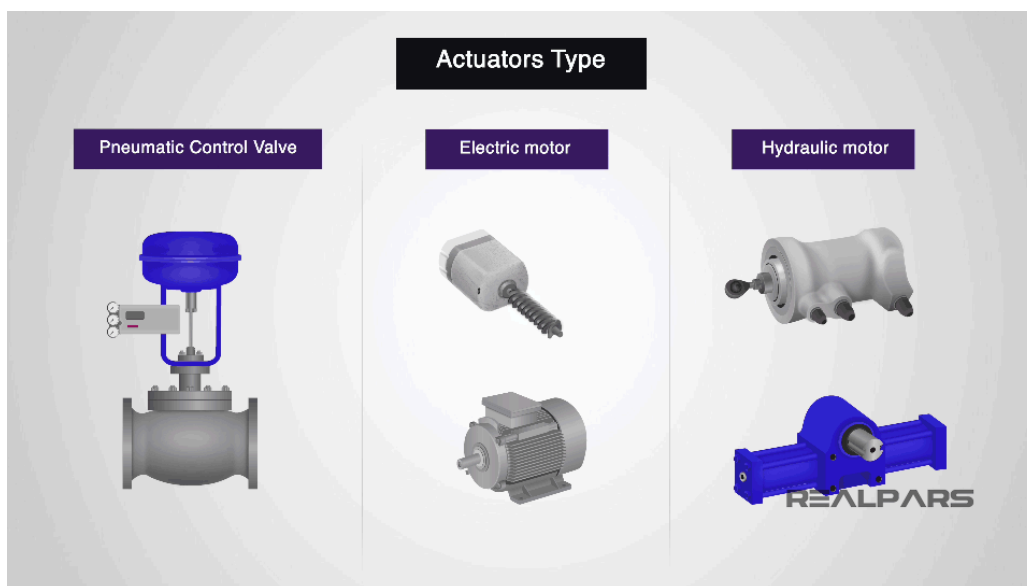
- **Electric actuators** use some form of electric energy to operate.
- **Hydraulic actuators** use a variety of liquids as a source of energy.
- **Pneumatic actuators** are operated by compressed air.



## Types of Actuators

Typical actuator types in the industry include:

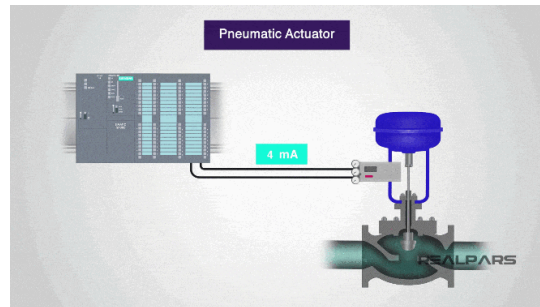
- Electric Motors
- Hydraulic motors
- Pneumatic Control Valves



## 4-20 mA Pneumatic Actuator

Let's look at a typical **Pneumatic Actuator** in action! The PLC analog output card produces a **4-20 mA** current to move the valve from fully open to fully closed.

The **4-20 mA** current will be converted to pneumatic pressure which becomes the source of energy to operate the actuator.



## Pneumatic Actuators

Pneumatic actuators can utilize a cylinder, diaphragm, or rotary vane design. Compressed air acts upon a piston or paddle to move it in one direction. A counterforce, consisting of a spring or a second volume of compressed air, is applied to the opposite side of the piston or paddle to reverse or maintain position. For industrial control, a pneumatic actuator requires positioning electronics, and an instrument air system. Of the three types of actuators, pneumatics tend to have the lowest initial price. However, operating costs are high due to the need to generate clean, dry, compressed air. Maintenance requirements are also the highest of the three actuator types.

## Hydraulic Actuator

Electro-hydraulic actuators operate a piston with pressurized oil. A motorized pump sends fluid from a reservoir through a control valve(s) to opposite sides of a cylinder. The high power density of this system allows for high thrust, and fast operating speeds. Electro-hydraulics tend to have a high initial price. Routine maintenance involves periodic replacement of seals, o-rings, etc. Since the oil used in hydraulic equipment can pose a fire hazard, this type of actuator may not be suitable for some industrial environments.

Electric actuators utilize an electric motor and gear reduction to produce a force or torque. A variety of technologies can be used for an electric actuator design. Motors may use AC or DC power, and an asynchronous (squirrel cage) or synchronous design. Gearing systems can include worm, spur, scotch yoke, among others. Gear lubrication may consist of an oil filled gear box, or heavy duty grease applied to the gear surfaces. Various accessories are often available to supervise and report on the actuator's status and operating conditions. There is a wide variety of electric actuators on the market with different technologies, prices, performance, and quality.

Because actuators are closely integrated with driven elements like valves,

and dampers, they are often referred to as assemblies. Examples include motor operated valves (MOV), air operated valves (AOV), and damper drives. Frequently, the actuator may be assumed to come with the equipment package. Examples include control valve, or air damper. In practice, actuators are the critical link between the control system and the final element. Its performance can significantly impact many aspects of an industrial plant's operations and products. Care should be taken to ensure the correct actuator technology and type is selected.

When selecting an actuator, the actuator should have performance characteristics that will enable a control system to perform as designed.

The key actuator performance characteristics are as follows:

- Precise, repeatable positioning typically better than 0.15% of span.
- The ability to start and stop instantaneously without dead time or position overshoot.
- Continuous duty rating without limitations on the number of starts per minute.
- Perform consistently and unaffected by load.
- Rugged industrial design capable of operating in difficult environments without an effect on performance.
- Minimal periodic maintenance required.

An actuator designed with these characteristics provides two extremely important advantages:

1. An ability to follow the demand signal from the controller precisely and instantly. This ensures that the actuator responds exactly as directed by the controller. Thus, the actuator is not the limiting factor in the control loop and the controller can function to its optimal levels.
2. A high degree of maintenance-free reliability. An actuator designed to function as outlined above by default is more rugged than typical actuators. By design, then, it is capable of a much higher degree of reliability.

## REFERENCES:-

## SUMMARY:-

An actuator is a **mechanical or electro-mechanical device that converts energy from a control signal into mechanical motion**. In simpler terms, an actuator is a device that allows controlled movement or positioning. Actuators need a control signal and a

source of energy to bring about mechanical motion.

**TOPIC:- Thermal/magnetic, mechanical actuators, soft actuators**

**OBJECTIVES:-** Thermal and magnetic actuators usually consist of shape memory alloys that can be heated to produce movement.

**OUTCOME:-** Applications for soft actuators include **artificial muscles, wearable technologies, medical devices, haptic devices, and soft grippers**. Manufacturing soft actuators that accurately mimic biological systems will provide robots with superior human and animal-like physical behaviors.

**Thermal and magnetic, mechanical actuators, soft actuators**

Thermal actuators



Thermal actuators.

Thermal actuators produce a linear movement – or stroke – as a result of the expansion and contraction of a thermally sensitive material within them.

### **Applications**

Integrating temperature sensing and actuation into a single device, thermal actuators are commonly used in applications such as temperature control, fluid mixing and diverting, freeze protection and high-temperature shutoffs.

Thermal actuators are the key operative components inside devices ranging from thermostats to valve products like thermal bypass valves, thermostatic mixing valves, tempering valves and thermostatic

balancing valves.

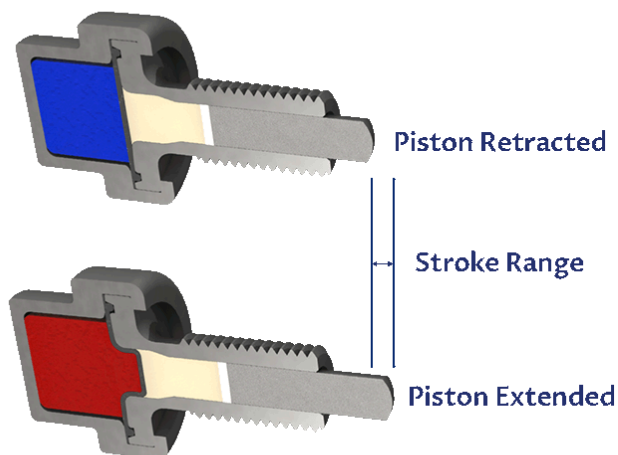
A thermostatic actuator acts as a valve actuator to open or close a valve based on a temperature setpoint to, for example, continuously regulate the flow of hot water to a radiator heating a room.

A thermal actuator combined with a source of heat such as a positive temperature coefficient thermistor is known as a wax motor, commonly used in appliances such as washing machines or dishwashers to engage locks or operate dispenser latches.

### Operating principle

Thermal actuators convert thermal energy into kinetic energy, or motion. A typical thermostatic actuator consists of a temperature-sensing material sealed within a housing by a diaphragm that pushes against a plug to move a piston within a guide.

#### **“Cold Position” - Solid State**



#### **“Hot Position” - Liquid State**

Thermal actuators produce a movement – or stroke – as thermally sensitive material within them expands and contracts due to changing temperature. Source:

The key element of a thermal actuator is the temperature-sensing material. It is this material that expands and contracts based on the temperature of the device, causing the piston to move. This material could be a liquid, gas, wax-like substance or other material that changes volume based on temperature.

In the case of wax materials, the substance is carefully engineered to undergo a phase change within a specific temperature range. When the temperature rises past the melting point, the wax changes phase from

solid to liquid, expanding in volume significantly. This expansion pushes against a diaphragm containing the wax, which forces a piston to move within a guide.

When the temperature falls back below the melting point, the wax changes phase back to solid from liquid. This results in a contracting volume of material which relaxes pressure on the diaphragm. A spring causes the piston to retract back within the guide to its original position.

Typical stroke ranges for wax-based thermal actuators range from 0.05 in to 0.5 in , generating pushing forces from 5 lb. to 75 lb. within temperatures ranging from -150° F to 300° F.

## Advantages



A thermostatic mixing/diverting temperature control valve is an all-in-one fluid proportioning solution that automatically mixes or diverts fluids based on temperature with no external components needed. Source: In many applications, thermal actuators have a number of advantages compared to traditional solutions.

For instance, the thermal actuator inside of a thermostatic or thermal valve enables it to operate as a single device that integrates temperature sensing, actuating and fluid control functions. Compared to a conventional setup that employs a solenoid valve and electronic sensing system, a thermal valve eliminates a number of external components.

The thermal actuator removes the need for the solenoid, temperature sensor such as a thermocouple , electronic instrumentation and control circuitry, and power supply such as a battery or DC power supply .With fewer components, the thermal valve is easier and cheaper to install. The overall design complexity and total cost of the solution are reduced.

Since the thermal valve does not require a supply of electricity, the device is

immune to power failures and dangerous events like short circuits. Reliability and safety are thus improved over a traditional setup.

Thermal actuators are compact units with a high ratio of power to size and weight. They do not require periodic calibration and offer safe, reliable operation in hazardous or extreme applications.

### Mechanical actuators



The mechanical actuators are devices that transform the rotary pressure movement to the entrance, in a linear movement in the exit.

The mechanical actuators are applicable for the fields where require linear movements such as: elevation, traslación and linear positioning.

Some of the advantages that offer the mechanical actuators are: high reliability, simplicity of utilisation, minimum manutención, hygiene and precision of positioning; irreversibilidad according to the model of application and synchronism of movement.

Inside the field of the mechanical actuators exist two types of movement: linear mechanical/ actuators with husillo traslante (series ST, M type1) (B2 type1) and linear mechanical/ actuators with husillo rotante. (Series SR, series M type2) (series BL type2).

PMZ Comatrans, S.A. has the exclusividad to offer the 2 possible systems of mechanical performance: mechanical actuators endless tops and mechanical actuators conical gears.

The range of mechanical actuators besides comprises different series with

finishings in smelting or aluminium, with different formats of fixation.

The technical capacity of the personnel of PMZ Comatrans, S.To it puts of self-evident in the search of solutions and proposal for all type of application like the mechanical actuators for solar power.

The mechanical actuators Type ST and SR in aluminium possess big variety of sizes and accessories to have a perfect control on the trip.

From R1/10 to R1/30.

High mechanical actuators loads, mechanical cats of system of helicoidal conical gears thought for high speeds and high loads (R1/1 to R1/5).

### Magnetic Actuators

A new family of linear voice coil actuators provides increased safety and long service life for valve control applications. The devices feature two proprietary spring designs based on magnetic, rather than mechanical technology, that ensures constant and precise valve position in either the open or closed setting when electrical power is off.



These “contact-less” actuators user fewer parts and eliminate friction, a problem associated with wear-related performance of mechanical springs. Each actuator incorporates either a magnetic extension or compression spring inside a coil assembly. In effect, the entire actuator is an electromagnetic spring controlled by the current in the winding. Both magnetic spring designs also include stops, which limit the position of the moving coil when it is not energized. When the coil is de-energized, it remains in a set, fixed position.

The field of Soft Robotics promises many potential benefits such as resistance to mechanical damage, contact compliance, and better compatibility for human-robot interaction. However, realization of this potential has been hindered by the lack of a soft actuator. While flexible batteries and circuits exist, a key challenge for soft robotics remains the

absence of an electrically-driven motor made entirely of soft materials, capable of delivering large strains. Here we propose a new meta-material that produces strain density that is more than 10 times better than natural muscle. We demonstrate the material in a range of robotic applications. We suggest that this new material opens the door to fully-soft robots.

We aimed to synthesize a cheap, simple, user- and environment-friendly material comprised of, preferably, food-safe and bio-compatible materials. We chose PDMS-based silicone elastomer, a non-hazardous elastomer widely used for soft robotic applications, as a matrix material, and ethanol, a widely used alcohol with boiling temperature 78.4 °C and matrix-compatibility, as the active fluid .

## **REFERENCES:-**

## **SUMMARY:-**

An actuator is a device that converts energy into physical motion, and most actuators produce rotary or linear motion. The three major types of actuators are hydraulic, pneumatic, and electric, and picking the best one depends on what kind of robotic structure you are designing.